AgriChain

Decentralized Crop Supply Chain & Traceability on Botchain

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1. Executive Summary

AgriChain is a decentralized crop supply chain and traceability platform deployed on Botchain. It enables farmers to register produce lots on-chain, stakeholders to append verifiable checkpoints throughout the supply journey, and buyers to trace the complete provenance of any agricultural product with full transparency and immutability.

Africa's agricultural supply chain suffers from opacity, counterfeit produce labeling, inefficient logistics, and a lack of trust between rural farmers and urban buyers. AgriChain addresses these problems by anchoring every step of the supply chain — from field registration to final sale — on the Botchain public ledger.

2. Problem Statement

2.1 Supply Chain Opacity

In Africa, over 40% of agricultural produce is lost between farm and consumer due to poor logistics visibility and lack of real-time tracking. Buyers cannot verify whether produce was grown organically, handled safely, or stored correctly.

2.2 Counterfeit Labeling

Fraudulent quality grades and false origin claims are rampant in wholesale markets. Verification currently relies on fragile paper-based documentation that is easily altered.

2.3 Farmer Disempowerment

Smallholder farmers lack credentialed records of their produce history, preventing them from accessing premium markets or proving quality to international buyers.

3. Solution – AgriChain Architecture

AgriChain deploys a single Solidity smart contract on Botchain that manages three core primitives: Farmer Profiles, Produce Lots, and Supply Chain Checkpoints.

3.1 Smart Contract Components

Component	Description	On-Chain Data
Farmer Registry	Stores verified farmer identities	Name, location, lot count, verified flag
Produce Lots	Immutable lot records per harvest	Crop, weight, price, status, owner
Checkpoints	Append-only audit trail per lot	Handler, location, action, timestamp
Marketplace	Native BOT token lot purchases	Buyer, sale price, platform fee

3.2 Verification Flow

- Farmer registers on-chain with name and location
- Platform verifier approves the farmer wallet
- Farmer creates a produce lot (crop type, weight, harvest date, price)
- Each handler adds a checkpoint (quality check, loading, transit, warehouse)
- Buyers browse the marketplace and purchase lots directly in BOT
- Ownership transfers atomically; platform fee deducted automatically

4. Milestones & Activities

Milestone	Activity	Timeline	Deliverable
M1 – Contract Deploy	Deploy AgriChain.sol on Botchain testnet; unit tests	Month 1	Verified contract address
M2 – Farmer Onboarding	Onboard 50 pilot farmers; KYC via verifier multisig	Month 2	50 verified farmer profiles
M3 – Frontend Launch	Launch index.html dApp; wallet connect integration	Month 2–3	Live dApp on Botchain
M4 – Marketplace Live	Enable lot listings and BOT-native purchases	Month 3	First 10 lot sales
M5 – Mobile QR Trace	QR code per lot for consumer scanning (scan-to-trace)	Month 4	QR traceability MVP
M6 – Oracle Integration	Integrate weather and logistics oracles for auto-checkpoints	Month 5–6	Oracle-driven checkpoints
M7 – Mainnet Launch	Full mainnet deploy; marketing to agri-cooperatives	Month 6	Production launch

5. Tokenomics & Economics

AgriChain operates on native BOT. Platform fees (default 1%, adjustable up to 5%) are collected on every lot purchase and flow to the contract owner treasury. No additional token is issued; the protocol is fee-based and self-sustaining.

- Platform Fee: 1% of each lot sale in BOT
- No pre-mine or VC allocation — revenue from usage
- Farmer verification is free (subsidised by platform fees)
- Future DAO governance via multisig upgrade path

6. Technical Stack

Layer	Technology	Purpose
Smart Contract	Solidity ^0.8.20 on Botchain	Core logic, state, payments
Frontend	HTML5 / CSS3 / Vanilla JS	Farmer and buyer interface
Evidence Storage	IPFS (CID references on-chain)	Quality reports, images
Wallet	Botchain-compatible wallet	Transaction signing
Oracle (roadmap)	Botchain oracle / Chainlink	Logistics and weather events

7. Team & Governance

AgriChain is developed under the Botchain Agriculture Series initiative. The initial contract owner is a 2-of-3 multisig between the core developer, a regional agri-cooperative representative, and a technical advisor. Platform fee parameters and verifier lists are owner-controlled, with a roadmap to on-chain DAO governance by Q3 2027.

8. Conclusion

AgriChain leverages Botchain's speed and low fees to make immutable supply chain traceability affordable for smallholder farmers across Africa. By anchoring every crop lot on-chain from harvest to sale, AgriChain builds the trust infrastructure needed for agricultural commerce to thrive in a digital economy.